

Lab 11

Data Structures with AI: Implementing Fundamental Structures

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Task 1 – Stack Implementation

Code:

```
class Stack:
    def __init__(self):
        self.items = []
    def push(self, item):
        self.items.append(item)
    def pop(self):
        if not self.is_empty():
            return self.items.pop()
        return "Stack is empty"
    def peek(self):
```

```

if not self.is_empty():
    return self.items[-1]
return "Stack is empty"
def is_empty(self):
    return len(self.items) == 0

```

Task 2 – Queue Implementation

Code: class

Queue:

```

def __init__(self):
    self.items = []
def enqueue(self, item):
    self.items.append(item)
def dequeue(self):
    if self.items:
        return self.items.pop(0)
    return "Queue is empty"
def peek(self):
    if self.items:
        return self.items[0]
    return "Queue is empty"
def size(self):

```

```
return len(self.items)
```

Task 3 – Singly Linked List

Code: class

Node:

```
def __init__(self, data):
```

```
self.data = data self.next =
```

None class LinkedList:

```
def __init__(self):
```

```
self.head = None
```

```
def insert(self, data): new_node
```

```
= Node(data)
```

```
if not self.head:
```

```
self.head = new_node
```

```
return temp = self.head
```

```
while temp.next: temp
```

```
= temp.next
```

```
temp.next = new_node
```

```
def display(self):
```

```
temp = self.head
```

```
while temp:
```

```
print(temp.data,
```

```
end=" -> ") temp
= temp.next
print("None")
```

Task 4 – Binary Search Tree (BST)

Code: class

BST:

```
def __init__(self, key):
    self.key = key
    self.left = None
    self.right = None
def insert(self, value):
    if value < self.key:
        if self.left is None:
            self.left = BST(value)
        else:
            self.left.insert(value)
    else:
        if self.right is None:
            self.right = BST(value)
        else:
            self.right.insert(value)
def inorder(self):
    if self.left:
```

```

        self.left.inorder() print(self.key,
end=" ")
if self.right:
    self.right.inorder()

```

◆ Task 5 – Hash Table (Chaining) Code:

```
class HashTable:
```

```

    def __init__(self, size=10):
        self.size = size self.table = [[] for _ in
range(size)] def _hash(self, key): return
hash(key) % self.size def insert(self, key,
value): index = self._hash(key)
self.table[index].append((key, value))
def search(self, key): index =
self._hash(key) for k, v in
self.table[index]:
    if k == key: return
    v return "Key
not found" def
delete(self,
key): index =
self._hash(key)

```

```

        for i, (k, v) in
            enumerate(self
                .table[index]):
            if k == key:
                del self.table[index][i]
                return "Deleted"
        return "Key not found"

```

Task 6 – Graph (Adjacency List)

Code: class

Graph:

```

    def __init__(self):
        self.graph = {}
    def
add_vertex(self, vertex):
        if vertex not in self.graph:
            self.graph[vertex] = []
    def add_edge(self, v1, v2):
        self.add_vertex(v1)
        self.add_vertex(v2)
        self.graph[v1].append(v2)
        self.graph[v2].append(v1)
    def display(self):

```

```
for vertex in self.graph:  
    print(vertex, "->", self.graph[vertex])
```

Task 7 – Priority Queue

Code:

```
import heapq  
class  
PriorityQueue:  
    def __init__(self):  
        self.heap = []  
    def enqueue(self, priority, item):  
        heapq.heappush(self.heap, (priority, item))  
    def dequeue(self):  
        if self.heap:  
            return heapq.heappop(self.heap)  
        return "Queue is empty"  
    def display(self): print(self.heap)
```

Task 8 – Deque

Code:

```
from collections import deque  
class DequeDS:  
    def __init__(self):
```

```

        self.deque = deque()
    def insert_front(self, item):
        self.deque.appendleft(item)
    def insert_rear(self, item):
        self.deque.append(item)
    def remove_front(self):
        return self.deque.popleft()
    def remove_rear(self):
        return self.deque.pop()

)

```

Task 9 – Campus Resource Management System

Feature → Data Structure Mapping

Feature	Data Structure	Justification
Student Attendance complexity.	Hash Table Tracking	Fast lookup by student ID. $O(1)$ average time
Event Registration		Dynamic insertion and removal of Linked List participants

Library Book Borrowing BST access.		Books can be organized by ID for sorted
Feature	Data Structure	Justification
Bus Scheduling	Graph	Routes and stops form connected networks.
Cafeteria Order Queue	Queue	Orders must be served in FIFO order.

Implemented Feature: Cafeteria Order Queue

```

class CafeteriaQueue:
    """Manage student food orders using Queue."""

    def __init__(self):
        self.orders = []

    def place_order(self, student_name):
        self.orders.append(student_name)
        print(f"{student_name} placed order.")

    def serve_order(self):
        if self.orders:
            served = self.orders.pop(0)
            print(f"Serving {served}")
        else:
            print("No orders to serve.")

    def display_orders(self):
        print("Current Orders:", self.orders)

```

```
# Example cq =  
CafeteriaQueue()  
cq.place_order("Rah  
ul")  
cq.place_order("Anit  
a") cq.serve_order()  
cq.display_orders()
```

Task 10 – Smart E-Commerce Platform

Feature → Data Structure Mapping

Feature	Data Structure	Justification
Shopping Cart	Linked List	Dynamic addition/removal of products.
Order Processing	Queue	Orders processed in FIFO order.
Top-Selling Products	Priority Queue	Ranked by highest sales.
Product Search	Hash Table	Fast product lookup by ID.
Delivery Route Planning	Graph	Warehouses and locations form network paths.

Implemented Feature: Product Search (Hash Table)

```
class ProductSearch:  
    """Product lookup system using Hash Table."""
```

```
def __init__(self):  
    self.products = {}
```

```
def add_product(self, product_id, name):  
    self.products[product_id] = name
```

```
def search_product(self, product_id):  
    return self.products.get(product_id, "Product not found")
```

```
# Example ps = ProductSearch()
```

```
ps.add_product(101,  
"Laptop")
```

```
ps.add_product(102,  
"Mobile")
```

```
print(ps.search_product(101))
```

```
print(ps.search_product(200))
```